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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: Ethics The Medical Ethics Review Board of the University Medical Center Groningen (UMCG) evaluated the study protocol (no. 2018220) and concluded that the Dutch Medical Research Human Subjects Act did not apply.
Consent: Study population, recruitment, and consent To ensure a representative study population, participants were recruited through the aforementioned care institutions (various backgrounds, regions, living situations) based on inclusion and exclusion criteria.
Inclusion and Exclusion Criteria
Study population, recruitment, and consent To ensure a representative study population, participants were recruited through the aforementioned care institutions (various backgrounds, regions, living situations) based on inclusion and exclusion criteria.
Attrition
not detected.
Sex as a biological variable
Table 1 Characteristics of the three diagnostic study groups A: Participants characteristics DS (N = 292) DS + Q (N = 119) DS + AD (N = 113) p Sex (male) 51.7 48.7 61.1 0.135 Mean age+-SD (min.-max.) 47.4+-9.0 (30.0-73.0) 53.8+-7.7 (33.4-74.0) 57.3+-6.4 (37.2-70.0) < 0.001 ID level (baseline): mild moderate severe () 11.3 66.8 21.9 16.0 57.1 26.9 8.0 62.8 29.2 0.160 IQ-score available () (mean IQ-score+-SD) 21.2 (42.3+-11.6) 22.7 (44.2+-9.4) 15.9 (40.4+-11.9) 0.387 0.543 Living independently with family assisted facilities other () 1.7 12.7 85.6 0.0 0.0 12.6 86.6 0.8 0.0 10.6 86.7 2.7 0.066 Attending day-care () 93.5 89.9 82.3 0.003 Type DS: full trisomy translocation mosaicism unknown () 37.0 0.3 0.3 62.3 39.5 0.8 0.8 58.8 45.1 0.9 0.9 53.1 0.706 Depression: treated untreated () 2.7 0.0 3.4 0.0 8.0 0.0 0.051 Epilepsy: treated untreated () 3.8 0.3 9.2 1.7 20.4 2.7 < 0.001 Hypothyroidism: treated untreated () 41.4 1.0 46.2 0.8 56.6 0.9 0.106 Vitamin B12 deficiency: treated untreated () 4.1 0.0 8.4 0.0 5.3 0.0 0.214 Hearing problems: treated untreated () 28.8 15.1 34.5 19.3 38.1 25.7 0.006 Vision problems: treated untreated () 58.9 19.2 59.7 22.7 58.4 26.5 0.355 Sleep apnea: treated untreated () 4.1 2.4 4.2 5.0 6.2 4.4 0.536 Chronic pain: treated untreated () 10.6 3.1 13.4 5.0 18.6 2.7 0.211 Physical disability () 7.9 7.6 8.8 0.929 Dependent on wheelchair () 0.3 1.7 5.3 0.003 Swallowing problems () 10.6 14.3 21.2 0.020 Dental problems causing eatingdrinking problems () 10.3 13.4 14.2 0.458 Verbal communication: able no longer never () 93.2 0.3 6.5 90.8 1.7 7.6 90.3 5.3 4.4 0.013 Any psychoactive medication use () 18.8 33.6 54.9 < 0.001 - Antiepileptics (N03A,) 5.1 12.6 25.7 < 0.001 - Antipsychotics (N05A,) 8.6 13.4 23.9 < 0.001 - Anxiolytics (N05B,) 0.7 2.5 8.8 < 0.001 - Hypnotics and sedatives (N05C,) 1.0 0.8 3.5 0.142 - Antidepressants (N06A,) 7.9 18.5 24.8 < 0.001 - Antidementia (N06D,) 0.0 1.7 5.3 < 0.001 - Opiods (N02A,) 0.0 0.0 2.7 0.004 Number of psychoactive medications (N = 0 1 2 3 4 5) 81.2 14.0 3.8 0.3 0.7 0.0 66.4 19.3 10.9 0.0 3.4 0.0 45.1 23.9 19.5 5.3 4.4 1.8 < 0.001 B: Informants

characteristics DS (N = 534 informants) DS + Q (N = 217 informants) DS + AD (N = 203 informants) p Informants per participant (N = 1 N = 2 or N = 3) 35.6 47.3 17.1 34.5 48.7 16.8 36.3 47.8 15.9 0.997 Sex (men) 21.6 19.3 15.9 0.436 Caregiver: day-care residential nurse family other () 6.8 53.2 0.2 36.0 3.8 7.4 54.8 0.9 33.6 3.2 4.4 57.1 0.5 34.0 3.9 0.779 Years knowing participant (< 2 2-10 10-20 > 20 years) 6.6 37.0 14.0 42.5 5.5 36.9 18.9 38.7 11.3 38.4 12.8 37.4 0.116 Hours per week with participant (< 10 10-20 > 20 hours) 27.9 32.3 39.8 29.0 19.8 51.2 33.5 19.7 46.8 < 0.001 ID level refers to the highest level of functioning (baseline) before dementia-related decline occurred.

Subject Demographics

Age: Inclusion criteria: phenotypical diagnosis of DS, aged >=30 years, and a stable dose of psychoactive medications (if any).

Randomization

not detected.

Blinding

For IRR, the same interview was scored by two interviewers blinded to each others scores.

Power Analysis

not detected.

Replication

not required.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
Data were analyzed using SPSS version 26 (IBM Corp.) and StataSE version 14.2 (StataCorp LP).	SPSS		Suggestion: SPSS, RRID:SCR_002865(link)
	StataCorp		Suggestion: Stata, RRID:SCR_012763(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Statistics For population characteristics (Table 1), Pearsons chi-squared tests were applied to compare categorical data between groups.	chi-squared tests
ANOVA tests were used to compare normally distributed continuous data (age and IQ-scores) between groups.	ANOVA
A Kruskal-Wallis test was used to compare the number of psychoactive medications between groups (categorial, not normally distributed). p-values are reported with three decimals.	Kruskal-Wallis test
To compare individual item scores (frequency change and severity change) between the three groups, Kruskal-Wallis tests were used.	Kruskal-Wallis tests
The effect of sex (malefemale) and depression (presenceabsence) were studied using Mann-Whitney U tests on the entire study group of 524 participants (regardless of dementia status).	Mann-Whitney U tests
However, for analyses of frequency change and severity change per item and care burden change per section, the level of significance was adjusted according to the procedure described by Benjamini & Hochberg 37 to account for multiple testing.	Benjamini & Hochberg

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